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 Studierichting: Informatica
 Oefeningenreeks 1B nr. 39.7

Opgave: Bereken in $\mathbb{C}$ de vierdemachtswortel van -81 .

Oplossing:

$$-81 = (x + yi)^2$$

Stel $(a + bi) = (x + yi)^2$. Dan $(a + bi)^2 = (x + yi)^4 = -81$

$$-81 = (a + bi)^2 \Rightarrow a + bi = \pm 9i$$

$$\text{Als } \mathbf{a + bi = 9i} = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = 9 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -\frac{9^2}{4} \end{cases}$$

$$\lambda^2 - \frac{9^2}{4} = 0 \Rightarrow \lambda_{1,2} = \frac{\pm\sqrt{9^2}}{2} = \begin{cases} + : \frac{9}{2} = x^2 \Rightarrow x = \pm\frac{3}{\sqrt{2}} = \pm\frac{3\sqrt{2}}{2} \\ - : -\frac{9}{2} = -y^2 \Rightarrow y = \pm\frac{3}{\sqrt{2}} = \pm\frac{3\sqrt{2}}{2} \end{cases}$$

$$\text{Dan is } x + yi \in \left\{ \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i, -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i \right\}$$

$$\text{Als } \mathbf{a + bi = -9i} = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = -9 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -\frac{9^2}{4} \end{cases}$$

$$\lambda^2 - \frac{9^2}{4} = 0 \Rightarrow \lambda_{1,2} = \frac{\pm\sqrt{9^2}}{2} = \begin{cases} + : \frac{9}{2} = x^2 \Rightarrow x = \pm\frac{3}{\sqrt{2}} = \pm\frac{3\sqrt{2}}{2} \\ - : -\frac{9}{2} = -y^2 \Rightarrow y = \pm\frac{3}{\sqrt{2}} = \pm\frac{3\sqrt{2}}{2} \end{cases}$$

$$\text{Dan is } x + yi \in \left\{ \frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i, -\frac{3\sqrt{2}}{2}i - \frac{3\sqrt{2}}{2} \right\}$$

$$\mathbf{Oplossing:} \quad x + yi \in \left\{ \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i, -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i, \frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i, -\frac{3\sqrt{2}}{2}i - \frac{3\sqrt{2}}{2} \right\}$$

References